

Public Economics and Development

Week 2: Empirical methods

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Causal inference in social sciences: Quantifying causal relationships

Econometrics: Clarify quantitative, causal relationship between different variables

- *Does higher commodity prices reduce wars? (Caselli et al 2015; Dube, Vargas 2013)*
- *Does extractive colonial institutional backdrop hurt current institutional development? (Acemoglu et al 2001; Michalopoulos, Papaioannou 2016)*
- *Does negative shocks to economic productivity lead to revolutions (Brückner, Ciccone 2011)*

Treatment effect measures the size of impact these events have on sample of interest

- Identify an event that induces some shocks to individual behavior/societal outcome
- Find a suitably comparable sample (i.e. the control group and treatment)
- Compare outcomes for those affected by the event to those not affected

For empirical economics, we collect data from a suitably defined population and use various methods to estimate the effects of these events.

Correlation vs Causation: Does X cause Y ?

A **correlation** is a statistical measure describing how the two variables co-move

- It captures *any* type of statistical dependence that moves the two variables together: Causation, but also others too!
- Not causal 1: Y can cause X (reverse causality)
- Not causal 2: X and Y co-move because of Z that affects both (Omitted variable bias)

A **causal** relationship is identified from changes in Y due to *exogenous* changes in X

- Changes in X may be a combination of changes from X alone + other factors that co-move with X and affect Y (You do not want the latter)
- Randomized experiments: One way of isolating clean changes in X
- Econometrics: Concisely explains the relationship between X and Y variables in a single equation (Even when experiment is not feasible)

Treatment effect in mathematics

Mathematical notations

- Y_i : observed outcome for individual i
- $X_i = x$: Treatment assignment ($x = 1$ for treated, 0 for control group)
- $Y_i(x)$: Outcome for treated individual if $x = 1$, control group if $x = 0$.

Treatment effect: How much does outcome differ between treated and control group

$$TE = Y_i(1) - Y_i(0)$$

- With large number of i , we want an average treatment effect

$$ATE = E[Y_i(1) - Y_i(0)] = E[Y_i(1)] - E[Y_i(0)]$$

- Missing **counterfactual**: If i is treated, it cannot be in the control group
- We need to make judgement on what the counterfactual looks like w/ econometrics

Another problem: How to attribute changes to X

Can we fully attribute $Y_i(1) - Y_i(0)$ to X_i ?

- Accurate estimation of treatment effects of X_i relies on the answer to above question
- True (Exogenous): If treated & control group are similar on other dimensions except X_i
 - ▶ e.g. Random assignments (randomized control trials)
 - ▶ Simple ordinary least squares (OLS) method suffices
- Usually, this is false (Endogenous): Factors other than X_i could affect $Y_i(1) - Y_i(0)$

When does the endogeneity problems happen and how to deal with it?

- Omitted variables: Not taking into account factors that affect outcomes
- Selection into treatment: Those with certain trait more likely to be assigned treatment
- Reverse causality: Y affects X , then X affects Y , and so on..
- Need other methods to make treated and control groups comparable
 - ▶ Panel regression, Instrumental variable, difference-in-differences, regression discontinuity

Reconciling potential outcomes and regressions

Potential outcomes provides framework to match observations to treatment effect

$$\begin{aligned}Y_i &= Y_i(1)X_i + Y_i(0)(1 - X_i) \\ &= Y_i(0) + X_i(Y_i(1) - Y_i(0))\end{aligned}$$

In terms of conditional expectations,

$$\begin{aligned}E[Y_i|X_i] &= E[Y_i(0) + X_i(Y_i(1) - Y_i(0))|X_i] \\ &= E[Y_i(0)|X_i] + E[Y_i(1) - Y_i(0)|X_i]X_i\end{aligned}$$

Generic regression setup between X_i and Y_i

$$\begin{aligned}Y_i &= \alpha + \beta X_i + u_i \\ E[Y_i|X_i] &= \alpha + \beta X_i + E[u_i|X_i]\end{aligned}$$

where u_i are errors (determinants of Y_i not captured by X_i)

Ordinary Least Square (OLS) and random assignments

Random assignment assumes that treatment is as good as random

- Only X_i is different across control and treated groups
- In mathematic terms, $E[u_i|X_i] = 0$

Reconciling with earlier notations, we get

$$\begin{aligned}E[Y_i|X_i] &= E[Y_i(0)|X_i] + E[Y_i(1) - Y_i(0)|X_i]X_i \\&= \alpha + \beta X_i + E[u_i|X_i] \\&= \alpha + \beta X_i\end{aligned}$$

- Here, estimate of β is the estimate of treatment effect of interest
- For accurate estimates, we can control for variables other than X_i as well
 - ▶ Helps address issues arising from omitted variables

Ordinary Least Square (OLS): Mathematical approach

Setup of OLS regressions

- Write

$$Y_i = \alpha + \beta X_i + u_i$$

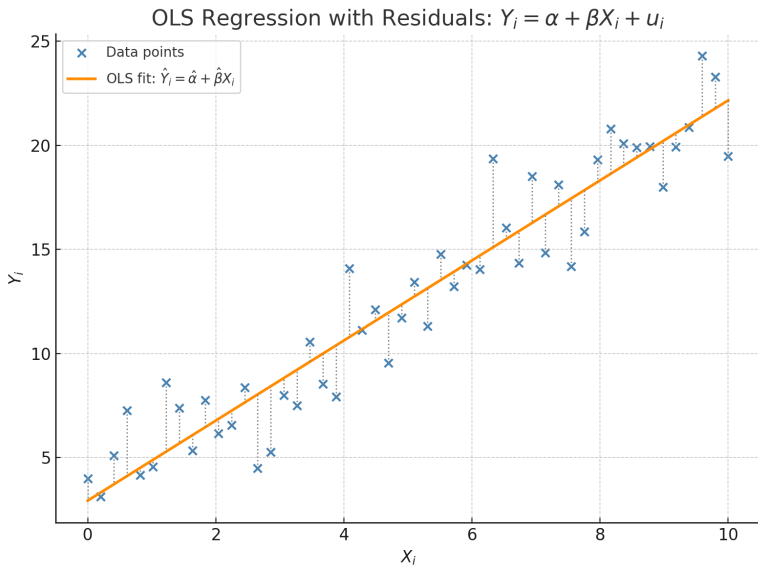
where X_i : independent variable and u_i is the error

- Exogeneity: We assume that u_i is uncorrelated with X_i , $E[u_i|X_i] = 0$ (very important!)
- OLS estimates the parameter of interest β by minimizing the sum squared of errors, giving us the β that best matches the patterns in the observation

$$\hat{\beta}_{OLS} = \min_{\beta} \sum_{i=1}^n (Y_i - \alpha - \beta X_i)^2$$

- We can do the same for α , which capture baseline average (average when all independent variables are zero)

Conceptualizing OLS



Internal and external validity problems: Why OLS is not always usable

External validity is concerned with the applicability to other contexts.

Internal validity assesses whether the model accurately represents the sample

- **Omitted variable bias:** Leave out factors affecting Y and are correlated with X ?
→ Multivariate regression, panel regression
- **Wrong functional form:** Not using proper functional form (quadratic, logs?)
→ add logs, squared variables, or use logit/probit
- **Measurement error:** If X does not have an accurate measure, estimates bias to 0
→ Instrumental variable
- **Sample selection bias:** certain groups more likely to be treated
→ Instrumental variable, Natural experiments
- **Simultaneous causality bias:** Y causes X too, leading to biased estimation
→ Instrumental variable

Most likely, we use natural experiment that uses external events/shock as treatment

When exogeneity fails: Mathematical explanation

$E[u_i|D_i] = 0$ rarely holds

- Treatment is often unrandom, or some are more likely to be treated than others
- This biases our results: Suppose that $E[u_i|D_i] = \phi D_i + v_i$ (u_i and D_i have correlation)

$$\begin{aligned}E[Y_i|D_i] &= \alpha + \beta D_i + E[u_i|D_i] \\ &= \alpha + (\beta + \phi)D_i + v_i\end{aligned}$$

With OLS on Y_i and D_i , $\hat{\beta}$ captures $\beta + \phi$ - Treatment effect and **endogeneity bias**

When and why do biases happen?

- Treated groups may be a good performer to begin with (and vice versa)
- Differences in baseline characteristics besides treatment → inaccurate TE estimates

Need to control for ϕ with individual/time indicators (Panel), exogenous proxy (IV), or natural experiments

Panel regression fixed effects approach

Setup

- We observe multiple individuals for multiple time periods ($i \in \{1, \dots, N\}, t \in \{1, \dots, T\}$)

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \dots + \beta_k X_{k,it} + u_{it}$$

- Problem: Unobserved individual characteristics and time trends could affect outcomes
 - ▶ Individual's unobserved ability/characteristics (e.g. culture)
 - ▶ A bad shock in year t that affects everyone (COVID years)

Panel regression allows us to capture these using individual and time fixed effects

$$Y_{it} = \beta_1 X_{1,it} + \dots + \beta_k X_{k,it} + \alpha_i + \lambda_t + u_{it}$$

- α_i : Dummy for each individual $i \in \{1, \dots, N\}$
- δ_t : Dummy for each time $t \in \{1, \dots, T\}$

Instrumental variables: Exogeneity amidst endogeneity

OLS hinges on independent variable X_i being uncorrelated with u_i

- This often breaks: omitted variable bias, measurement error, and reverse causality etc..
- Parts of X_i that are correlated with u_i : What we want is part of X_i independent of u_i .

Solution is to find another variable Z_i that are

- (1) Correlated with X (relevancy),
- (2) not correlated with u (exogeneity): This gets us part of X_i independent of u_i
- Exogeneity is difficult to argue (usually argue that Z_i affects Y_i only through X_i)

This is estimated with two stage least squares (2SLS)

- Original Equation: $Y_i = \alpha + \beta X_i + u_i$
- First stage: $X_i = \delta_0 + \delta_1 Z_i + v_i \rightarrow$ Get $\hat{X}_i = \hat{\delta}_0 + \hat{\delta}_1 Z_i$ ($\hat{X}_i$ is uncorrelated with u_i)
- Second stage: $Y_i = \beta_0 + \beta_1 \hat{X}_i + u_i$

Instrumental variables using natural environment

Brückner, Ciccone, "Rain and the Democratic Window of Opportunity", *Econometrica*, 79(3), 923-947

- Question: Can transitory shocks to income (X_i) lead to democratic changes (Y_i)?
- Income shocks can be endogenous: Places with higher income are likely more democratic to begin with
- Use (lack of) rainfall (Z_i): Correlates with income through agricultural productivity but unlikely to affect democratic transitions alone
- Plausible: Very hard for rainfall itself to lead to democratization
- Result: 1% transitory negative income shock $\rightarrow$ 1.3 percentage point rise in democratic transitions

Instrumental variables using natural environment

	Democratic Transition			Democratization Step		
	(1) LS	(2) 2SLS	(3) 2SLS	(4) LS	(5) 2SLS	(6) 2SLS
Log GDP, $t - 1$	0.056 (0.058)	-1.285** [0.027]		-0.053 (0.051)	-1.471** [0.029]	
Country specific recession, $t - 1$			0.235** [0.027]			0.279** [0.029]
Country fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Country time trend	Yes	Yes	Yes	Yes	Yes	Yes
Common time effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	700	700	700	867	867	867
First Stage for Log GDP per capita/Country Specific Recession, $t - 1$						
Log rainfall, $t - 1$		0.095*** (0.037)	-0.519*** (0.164)		0.094*** (0.032)	-0.494*** (0.151)
Country fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Country time trend	Yes	Yes	Yes	Yes	Yes	Yes
Common time effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	700	700	700	867	867	867

Natural experiments: Difference-in-differences

Assume two time periods (before and after) and two groups (treated and controlled)

- You have the following regression

$$Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 P_{it} + \beta_3 D_{it} P_{it} + u_{it}$$

where $D = 1$ if in treatment group, $P = 1$ if after treatment started

- You can get the following group averages

(1) Treated, After: $E[Y_{it}|D_{it} = 1, P_{it} = 1] = \beta_0 + \beta_1 + \beta_2 + \beta_3$

(2) Treated, Before: $E[Y_{it}|D_{it} = 1, P_{it} = 0] = \beta_0 + \beta_1$

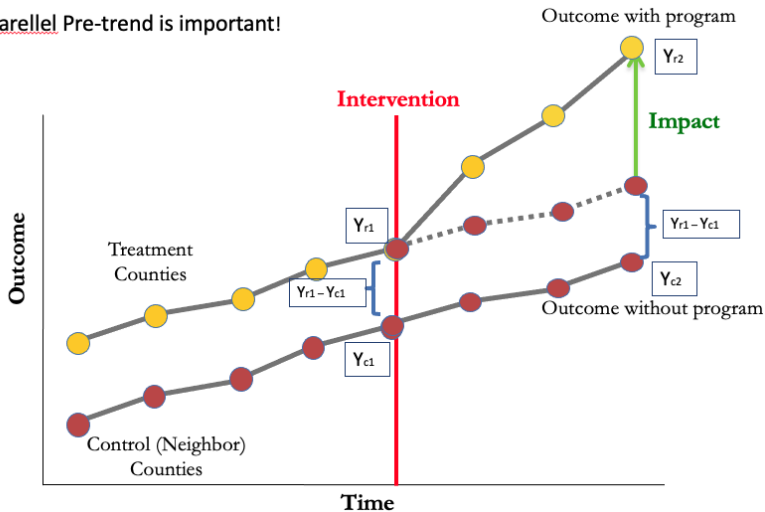
(3) Controlled, After: $E[Y_{it}|D_{it} = 0, P_{it} = 1] = \beta_0 + \beta_2$

(4) Controlled, Before: $E[Y_{it}|D_{it} = 0, P_{it} = 0] = \beta_0$

- DID: $[\bar{Y}_{\text{treat,after}}(1) - \bar{Y}_{\text{treat,before}}(2)] - [\bar{Y}_{\text{control,after}}(3) - \bar{Y}_{\text{control,before}}(4)] \rightarrow \hat{\beta}_3$
- Can be generalized to many time periods and many groups
 - ▶ Two-way fixed effects, event-studies

Natural experiments: Conceptualizing Difference-in-differences

§ Parallel Pre-trend is important!



Natural experiments: Difference-in-Differences

Andreas V. Jensen (2025) “Educating for Democracy? Going to College Increases Political Participation”, *British Journal of Political Science*, 55,1-12

- Question: Does going to college increase political participation?
- Method: Uses a survey of US students who graduated high school at 2004, observe differences in election participation in 2004 and 2008

$$\text{Turnout}_{it} = \alpha_i + \delta_t + \beta[\text{College}_i \times \text{Post}_t] + \gamma X_{it} + u_{it}$$

- Here α_i and δ_t are individual and time fixed effects
(Can you find their equivalent variables in the previous equation?)
- Treated: Those who entered college between 04-08
- Control: Did not go to college
- Result: College enrollment increase participation by 13 percentage points

Natural experiments: Difference-in-Differences

Table 1. Effect of College Education on Voter Turnout

	Full Sample		
	(1)	(2)	(3)
<i>Attended College</i> × <i>Post Period</i>	0.137*** (0.014)	0.130*** (0.016)	0.107*** (0.016)
Time FEs & Individual FEs	✓	✓	✓
Pre-college Voting × Time FEs	✓	✓	✓
Time-varying Controls		✓	✓
Additional Pre-college Covariates × Time FEs			✓
<i>Units</i>	10,426	9,652	9,554
<i>Observations</i>	19,813	18,298	18,118

Natural experiments: Regression discontinuity

Comparing individuals just entering the treatment vs marginally did not receive one

- Assume you are treated if $X_i \geq c$ and not if otherwise
- Then, run this regression on $X_i \in [c - h, c + h]$ for some small h , along with covariate Z_i

$$Y_i = \beta Z_i + \gamma(1[X_i \geq c] \times Z_i) + \delta(Z_i \times (X_i - c)) + \mu(Z_i \times (X_i - c) \times 1[X_i \geq c]) + u_i$$

For simplicity, assume that no other independent variables are in place: $Z_i = 1$

- ▶ $1[X_i \geq c] = 1$ if i passes threshold for eligibility (0 if otherwise)
- ▶ γ captures the treatment effect: Difference between those in and out of treatment
- ▶ We also allow slope to be different under and above the cutoff: δ for people below cutoff and $\delta + \mu$ for those above

Caveats

- Treatment effect estimated here is local: May not hold for out-sample
- There may be cases where threshold is not strict (fuzzy RD)

Regression discontinuity in use with electoral rules

Thomas Fujiwara (2011) "A Regression Discontinuity Test of Strategic Voting and Duverger's Law", *Quarterly Journal of Political Science*, 6, 197-233

- Question: Do third-party candidates perform well in majoritarian elections with run-offs than ones with no run-offs?
- Design: In Brazil, precincts with more than 200,000 registered voters have run-off elections for mayors (otherwise, no run-offs)
- Findings: Third-party candidates perform significantly worse in places without run-offs, suggesting that voters consider winning probabilities of candidates into account

Natural experiments: Regression discontinuity

